

PODGORICA GOLUBOVCI, Montenegro

WMO: 134620

Lat: 42.367N

Lon: 19.250E

Elev: 108

StdP: 14.64

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.1	29.6	8.1	8.3	35.4	12.5	10.5	37.4	29.9	36.6	27.2	39.1	6.6	0	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	97.3	70.6	94.8	70.2	92.2	69.7	73.9	89.5	72.5	88.5	71.3	87.4	6.8	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.1	107.5	80.7	67.7	102.0	78.3	66.2	97.1	77.5	37.4	89.3	36.1	88.4	35.1	87.8	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.3	19.9	16.4	22.3	100.9	3.0	3.6	20.1	103.5	18.4	105.6	16.7	107.6	14.4	110.3		
			20.2	76.8	3.6	2.8	17.6	78.8	15.5	80.5	13.4	82.0	10.8	84.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	61.0	42.2	44.9	51.5	59.0	66.9	74.9	80.8	81.0	71.1	62.1	52.4	44.4
	DBStd	14.60	5.61	6.12	5.78	5.40	5.51	5.98	4.91	5.15	5.51	5.35	6.07	5.74
	HDD50	695	246	159	51	4	0	0	0	0	0	1	45	189
	HDD65	3074	706	563	418	189	45	5	0	0	10	119	380	639
	CDD50	4719	5	16	98	275	523	747	955	960	634	374	116	16
	CDD65	1624	0	0	0	10	103	303	490	496	194	28	0	0
	CDH74	17052	0	1	1	98	788	3068	5859	5683	1406	147	1	0
	CDH80	7673	0	0	0	14	179	1241	2932	2850	447	10	0	0
Wind	WSAvg	5.4	5.5	6.1	6.7	5.1	5.3	5.2	5.9	5.4	5.1	4.4	4.6	5.5
Precipitation	PrecAvg	67.2	6.7	6.2	5.6	5.8	3.5	2.2	1.8	2.6	5.3	7.0	9.5	8.5
	PrecMax	92.9	15.0	15.2	13.7	13.4	9.1	6.4	5.5	10.8	15.4	20.6	25.2	16.3
	PrecMin	38.9	0.0	0.0	0.2	0.6	0.0	0.3	0.0	0.2	0.0	0.0	0.8	1.4
	PrecStd	12.2	4.5	3.9	3.2	3.0	2.2	1.6	1.2	2.4	4.1	4.9	4.8	4.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.0	65.0	72.6	81.6	87.5	96.0	100.2	101.6	93.4	81.4	70.8	61.4
		MCWB	52.4	52.8	55.2	61.7	66.9	71.9	70.9	71.2	68.5	65.1	60.2	55.4
	2%	DB	56.0	60.7	68.1	76.6	84.6	92.9	97.4	98.1	88.7	78.0	67.2	58.5
		MCWB	50.0	51.1	54.1	59.7	66.2	71.2	70.7	70.7	67.1	63.5	58.9	53.7
	5%	DB	53.5	57.7	65.0	73.0	81.8	89.9	94.9	95.1	84.9	74.9	64.6	56.2
		MCWB	48.0	49.6	52.7	57.7	64.9	70.2	70.3	69.8	66.4	62.2	57.2	51.5
	10%	DB	51.1	54.8	61.8	69.8	78.6	87.0	92.2	92.5	81.8	72.1	61.8	53.7
		MCWB	46.6	48.1	51.1	56.8	63.8	69.0	69.7	69.5	65.7	61.3	56.0	49.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.0	56.1	60.6	63.6	69.7	76.0	76.4	75.7	71.8	68.4	62.9	58.2
		MCDB	57.2	60.5	65.1	77.2	83.9	90.2	91.4	91.5	83.3	76.1	66.8	59.5
	2%	WB	52.0	53.2	56.7	61.6	67.5	73.3	73.8	73.5	70.2	66.3	60.9	55.4
		MCDB	54.5	58.2	63.5	73.6	80.8	88.7	89.5	89.4	81.3	73.2	65.3	57.5
	5%	WB	49.9	51.4	54.7	59.8	66.1	71.6	72.4	72.1	68.8	64.6	59.0	53.0
		MCDB	51.9	55.8	61.1	69.5	78.9	87.1	89.0	88.4	79.7	70.9	63.1	55.1
	10%	WB	48.0	49.6	53.1	58.4	64.6	70.1	71.2	70.9	67.5	63.1	57.0	50.3
		MCDB	49.9	53.5	58.9	66.7	76.2	84.4	88.2	87.3	77.9	69.0	60.6	52.8
Mean Daily Temperature Range	5% DB	MDBR	13.4	14.4	15.2	16.4	17.2	18.3	19.3	19.4	17.0	16.3	14.9	13.5
		MCDBR	16.9	19.7	20.8	22.3	21.8	21.4	22.6	22.7	20.7	20.4	18.2	16.3
		MCWBR	11.1	11.7	10.6	9.7	8.9	8.3	7.5	7.4	7.9	9.6	10.5	10.6
	5% WB	MCDBR	12.5	15.2	16.9	19.4	19.9	20.5	20.5	20.8	18.0	16.2	14.8	10.6
		MCWBR	9.4	10.0	9.8	9.2	8.8	8.6	8.2	8.0	8.1	8.4	9.6	8.2
Clear-Sky Solar Irradiance	taub	0.339	0.365	0.415	0.471	0.460	0.442	0.443	0.454	0.431	0.398	0.368	0.339	
	taud	2.392	2.294	2.183	2.056	2.153	2.258	2.251	2.213	2.263	2.334	2.369	2.402	
	Ebn at Noon	251	261	260	254	260	265	263	256	253	247	237	239	
	Edh at Noon	26	33	42	51	47	43	43	43	38	31	26	23	
All-Sky Solar Radiation	RadAvg	518	748	1114	1458	1818	2108	2179	1944	1386	960	602	463	
	RadStd	62	114	157	163	149	144	125	130	115	102	65	87	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	-184	N/A	N/A

Nomenclature: See separate page